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Seismo ionospheric anomalies possibly associated with the 2018 M_w 8.2 Fiji earthquake detected with GNSS TEC

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Abstract

The detection of earthquake (EQ) related ionospheric anomalies using the Global Navigation Satellite System (GNSS) has emerged as a convincing approach and new field in the search for seismic precursors. Total Electron Content (TEC) from International GNSS services (IGS) network may help to distinguish seismo ionospheric anomalies from geomagnetic storm induced anomalies. In this paper, spatial and temporal TEC anomalies are investigated in 1 sec resolution data from GNSS stations around the epicenter of the August 19, 2018, Fiji EQ (M_w 8.2). Moreover,

median and standard deviation based anomaly detection techniques are implemented on TEC and the variations are quantified on the basis of z-test at $\alpha=0.05$ significance level. We find evidence of significant ionospheric anomalies of low intensity within 5-10 days before and 5 days after the main shock by estimating TEC from four GNSS stations within the EQ preparation period. On the other hand, a significant ionospheric anomaly of high intensity occurred after the main shock more than 5 days after the EQ, which is associated with an intense geomagnetic storm ($Dst = -150$ nT and $Kp \geq 7$). More analysis reveals that EQ induced TEC anomalies are significant during Local Time (LT) = 08:00-16:00 (for Fiji, LT= UT+12) and similarly the storm ($Dst = -150$ nT and $Kp \geq 7$) induced TEC anomaly prevails in all GNSS stations during LT= 08:00-18:00. The TEC intensity of the storm anomaly is more substantial in all four GNSS stations compared to the EQ induced TEC anomaly. All these anomalies before and after the main shock during the EQ preparation period are advantageous towards differentiating seismic versus storm anomalies. Furthermore, this analysis aids in the developing hypothesis of lithosphere ionosphere coupling through TEC monitoring.

Key words: GNSS TEC; Earthquake; Lithosphere-ionosphere coupling; Geomagnetic storm

1. Introduction

Many studies reported anomalous ionospheric and atmospheric perturbations associated with seismic events as detected with GNSS TEC and other atmospheric parameters during the EQ preparation period (Liu et al., 2006; Pulinets and Ouzounov, 2011; Shah et al., 2020a, b; Sekertekin et al., 2020). Additionally, seismic anomalies in the atmosphere and ionosphere before and after the main shock over the epicenters of EQs are detected using the ionospheric electron

content, F2 layer critical frequency (f_oF2) and other ionospheric and atmospheric indices from multiple other satellites and Remote Sensing procedures (Hayakawa et al. 1993; Pulinets and Boyarchuk, 2004; Ahmad et al., 2018; Tariq et al., 2019a; Shah et al., 2019b). Further analysis of seismo ionospheric anomalies include a more statistical approach (Liu et al., 2004; Shah and Jin, 2015), modelling the evolution and propagation of ionospheric anomalies over epicenter (Namgaladze et al., 2009; Kuo et al., 2011), and detections from multiple ionospheric satellite missions (Marchetti et al., 2019; Shah et al., 2019a; De Santis et al., 2019). Some other reports proposed that EQ cannot be predicted (Geller et al., 1997; Rishbeth, 2007) and some even preferred the storm induced ionosphere anomalies over seismo ionospheric anomalies (Afraimovich and Astafyeva, 2008; Thomas et al., 2017).

There are many reports on the execution of different methods for the relationship and monitoring of EQs and ionospheric anomalies from ground and space measurements during seismic preparation period. For example, abrupt enhancement in TEC from the worldwide GPS network is reported within 5 days before and after the main shock in a statistical analysis by Shah and Jin, (2015) and Shah et al., (2020c). Furthermore, Liu et al. (2001) established a substantial correlation of ionospheric anomalies and the Taiwanese Chi-Chi EQ in 1999 in a statistical analysis and emphasized the importance of a statistical approach for monitoring seismo ionospheric anomalies. Apart from this, Le et al. (2011) studied different morphological characteristics of seismo ionospheric anomalies including enhancement, depletion, and occurrence rate of positive and negative ionospheric anomalies. Parrot et al. (2006) studied the variations of nighttime electron density measurements from the DEMETER satellite and observed a sharp enhancement corresponding to the same time as the seismo ionospheric anomaly. Moreover, De Santis et al.

(2019) reported ionospheric anomalies within two months before to a few days before the main shocks of globally distributed EQs from Swarm satellite data.

There are many hypotheses for the explanation of lithosphere and ionosphere coupling around the epicenter of future main shock during the EQ preparation period (Kamogawa et al., 2006; Freund et al., 2009; Pulinets and Ouzounov, 2011; De Santis et al., 2015). For example, Pulinets and Boyarchuck (2004) explained that Radon discharge from the infected region before the main shock is responsible for the ionization of the nearby atmosphere and then ionosphere. Similarly, Freund et al. (2009) explained that p-holes phenomena occurred in stressed rocks inside the Earth as a result of tectonic forces and raised to the lithosphere. As a result, many reactions take place like ion molecular reaction, ion generation, and increment and decrement in the upward/downward electrical field around epicenter of future EQ. As more and more p-holes reach the surface a threshold is reached where which they are propagated to the ionosphere at different altitudes via the atmosphere (Shah et al., 2020a). In another model, Kou et al. (2014) performed a numerical analysis to describe the relationship of ionospheric anomalies with EQ by considering the variations in electrons around the epicenter. They numerically suggested that an electric field is established by the production of positive charge carriers around the EQ preparation zone which serve as carriers to transform the vertical charge flow to atmosphere and consequently drive plasma bubbles to the ionosphere.

In this paper, seismo ionospheric anomalies are investigated by TEC obtained from IGS permanent GNSS receiver stations within the vicinity of the M_w 8.2, 2018, Fiji EQ. The main objective of this paper is to investigate TEC anomalies within 5-10 days before and after the main shock and another objective is to find the difference between seismic and geomagnetic storm anomaly. Furthermore, different morphological characteristics of TEC anomalies associated with the EQ epicenter are

discussed, such as occurrence rate and spatial and temporal evolution. The manuscript is distributed into section 2, which presents the data and methodology, section 3, which provides the results and discussions, and section 4, which summarizes the conclusions.

2. Data and methodology

In this paper, ionospheric anomalies from TEC acquired from GNSS stations of IGS are studied by spatial and temporal analysis before and after the M_w 8.2, Fiji EQ. According to the United States Geological Survey (USGS) reports, Fiji EQ occurred at UT=16:25:06 at geographical location of Latitude (18.113°S) and Longitude (178.153°W), as in Figure 1. The EQ hit the region at 286 km North-East of the Ndoi Island of Fiji on August 19, 2018. Moreover, a total of 479 $M_w \geq 5.0$ foreshocks and aftershocks occurred around the epicenter of the M_w 8.2 Fiji EQ in 2018 (Fig. 1). Within this seismic sequence, 280 including M_w 7.1 were aftershocks, while others were foreshocks. All information about the EQs was retrieved from USGS via the online database (<https://earthquake.usgs.gov/earthquakes/eventpage/us1000gcii/executive>). This EQ (M_w 8.2) occurred as a result of normal faulting at > 100 km from the Tonga Trench in the South Pacific Ocean. This seismic activity is located where the Pacific plate moves towards the west of the Australian plate at a velocity of nearly 81 mm/yr. More than 9 EQs of $M_w > 7$ have occurred at this location over the past century (<https://earthquake.usgs.gov/earthquakes/eventpage/us1000gcii/executive>).

To differentiate EQ and storm anomalies in the diurnal TEC time series, the analysis of geomagnetic storm indices has equal importance (Arslan et al., 2019; Tariq et al., 2019b). Therefore, it is compulsory to examine the thermospheric variations in the ionosphere to isolate storm time variation in TEC. We study different indices of geomagnetic storms (K_p , Dst and F10.7) to investigate the storm time anomalies in TEC before and after the main shock (Fig. 2).

The data of geomagnetic storm indices are obtained from NASA's OMNI web (<https://data.nasa.gov/SpaceScience/OMNIWeb-Plus/>) and the Kyoto university space weather portal (<http://wdc.kugi.kyoto-u.ac.jp>).

For the analysis of TEC anomalies associated with EQ, it is mandatory for a GNSS station to operate within the EQ preparation zone and this procedure is used by previous studies (Marchetti et al., 2019; Shah et al., 2019a; De Santis et al., 2019, Shah et al., 2020c). The EQ preparation zone can be estimated around the epicenter of the impending main shock by the formula of Dobrovolsky et al. (1979). For any EQ, the seismogenic zone around the future epicenter can be defined as,

$$\mathcal{R} = 10^{0.43M} \text{ km}, \quad (1)$$

Where $\mathcal{R}$ is the radius of preparation zone in km and M is the magnitude of the EQ. The radius of the selected EQ is about 3357.4 km (Fig. 1). The radius of the critical region depends on the magnitude of the future EQ (Shah et al., 2020b). For example, a high magnitude EQ has a significantly large preparation zone and a low magnitude EQ has less preparation zone.

We study diurnal TEC series from the GNSS stations operating within the EQ critical region. The daily TEC (1 sec temporal resolution) are extracted from the RINEX files with a cut off angle $< 15^\circ$ (<ftp://data-out.unavco.org/pub/rinex/>).

For monitoring TEC anomaly, Slant TEC from every receiver is obtained from the amount of free electrons in a 1×1 square meter tube along the travel path of the signal between the satellites and receiver. Slant TEC is measured in the unit of TEC and 1 TECU is equal to 10^{16} electron/m² (Li et al., 2014). Slant TEC from each GNSS station is calculated as,

$$STE C = \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)} (L_1 - L_2 + \lambda_1(N_1 - N_2) - \lambda_2(N_1 - N_2) + \epsilon) \quad (2)$$

$$STE C = \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)} (P_1 - P_2 - (d_1 - d_2) + \epsilon). \quad (3)$$

In above equations, f_1 and f_2 are the carrier phase frequencies of the GNSS signal. Similarly, P and L are the pseudo-range and carrier phase observation of the delay path of the GNSS signal, respectively. The wavelength and pseudo-range of the GNSS signal are λ and d , respectively, the ambiguity in the signal is N , and the instrumental bias is b . The random residual in the GNSS signal along the ray path is ϵ .

STEC is transformed to vertical TEC (VTEC) using the following function (Klobuchar, 1987):

$$VTEC = STEC \cdot \cos\left(\arcsin\left(\frac{R \cdot \sin Z}{R+H}\right)\right), \quad (4)$$

Where R and H ($H = 350$ km in this case for Klobuchar model) are the Earth's radius and altitude of the top layer of the ionosphere, respectively and Z is the satellite elevation angle for pierce point positioning (Feltens, 2003; Hernández-Pajares et al., 2009; Heki and Enomoto, 2013).

In this paper, a statistical confidence bounds method of mean and standard deviation (SD) is implemented on the normal TEC distribution. Daily TEC before and after the EQ is estimated by bounds, where the bounds for a single day are calculated from the mean and SD of 15 days before and after that day. Furthermore, the normal TEC distribution is observed at $\alpha=0.05$ (significance level) as a Z-distribution (Shah and Jin, 2015). For the monitoring of EQ and storm induced ionosphere anomalies, normal TEC within the confidence bounds are defined as,

$$\overline{TEC} - Z_{(df, 1-\alpha/2)} * SE < \sigma < \overline{TEC} + Z_{(df, 1-\alpha/2)} * SE. \quad (5)$$

The parameters $\overline{TEC}$ and σ are the mean and standard deviation, respectively. Similarly, SE is the standard error, which is equal to $\sigma/\sqrt{n}$ (n is the number of daily TEC values for a single day). Moreover, we use Z-distribution at $1-\alpha/2$ percentile to detect clear TEC anomaly (Shah and Jin, 2015). Z-distribution at $1-\alpha/2$ percentile provide the freedom to detect minute variations in TEC of 95 % confidence level. TEC beyond the upper confidence bound (UB) is considered as positive ionospheric anomaly and TEC lower than the lower bound (LB) is termed as negative ionospheric

anomaly. The deviation in TEC (dTEC) from both UB and LB is quantified to show anomalous ionospheric conditions. The dTEC for all the GNSS stations before and after the main shock can be calculated using the method of Shah and Jin (2017) and Shah et al., (2019b) such that

$$Deviation(dTEC) = \begin{cases} \frac{OV - X_{high}}{X_{high}} \leftrightarrow OV > X_{high} \\ 0 \leftrightarrow X_{high} - X_{low} \\ \frac{OV - X_{low}}{X_{low}} \leftrightarrow OV < X_{high} \end{cases} \quad (6)$$

This equation clearly highlights variations in TEC due to the observed signal and allows a comparison between EQ anomalies versus storm anomalies.

3. Results and Discussions

In this paper, temporal and spatial TEC anomalies from GNSS stations within the Dobrovolsky et al. (1979) region are studied within the 20 days before and 10 days after the main shock of the M_w 8.2, Fiji EQ. A severe storm ($Dst = -150$ nT and $Kp \geq 7$) also occurred 7 days after the main shock (Fig. 2) and significant TEC perturbations were recorded as a response to this storm. The storm indices remained quiet during 20 days before the main shock ($Dst > -50$ nT and $Kp < 3$) and low intensity TEC anomalies occurred before the EQ.

Ionospheric anomalies are investigated in temporal TEC values from 4 GNSS stations within the EQ preparation zone. The OUS2 station showed low intensity TEC anomalies on days -3 and -2 before the main shock as well as the storm induced high intensity anomalies of 8 and 1.4 TECU on days 7 and 8, respectively, after the main shock (Fig. 3). The low intensity TEC anomalies are EQ induced and are seen within 5-10 days before the main shock and within the 5 days after the EQ. On the contrary, the high storm activity ($Dst = -150$ nT and $Kp \geq 7$) induced significantly higher perturbations during the 6-8 days after the main shock as seen in Figs. 3-6. The weakness of the EQ induced anomalies detected at the OUS2 station may be due to the station's low latitude and large distance from the EQ epicenter. The station is also located near the lower edge of the

equatorial ionization anomaly, which may also explain the weak ionospheric perturbations. It is noteworthy that storm induced TEC anomalies have high intensity, which become more prominent in the calculation of dTEC and percentage deviation in Table 1. Similarly, the AUCK station recorded minute daytime TEC anomalies before the main shock and a significantly higher pattern of ionospheric anomalies in daytime and nighttime TEC which are associated with the storm (Fig. 4). It is also evident that geomagnetic perturbations are stronger than the seismic anomalies detected at the AUCK station. The high deviations in TEC anomalies induced by the storm detected by both the OUS2 (Fig. 3) and AUCK (Fig. 4) stations may be due to the stations' low latitude locations. Lin et al. (2007) also analyzed the impact of geomagnetic storms in the ionosphere at a low latitude western pacific region and suggested that geomagnetic storms cause high impact at low latitude regions during July-August. The KOUC station detected pre seismic (5 TECU on day -1) and post seismic (3 TECU on day +2) anomalies related to the EQ and this station is very close to the epicenter as well as equator (Fig. 5). In Figure 5, KOUC shows a huge TEC anomaly of 14 TEC due to the geomagnetic storm. As mentioned before, this may be due to the location of the station near the equator. The ASPA station is very near to the equator as well as epicenter of M_w 8.2 (Fig. 6) and it detected more anomalies related to both the EQ and storm. During the temporal analysis for dTEC in the second panel of Fig. 6, the EQ induced anomalies are between 5-10 TECU before and after the main shock, whereas the storm induced anomalies are > 10 TECU. The location of ASPA is close to the epicenter and therefore it detects a sequence of TEC anomalies because more energy propagates to the nearby station which possibly is situated over the fault lineaments; ASPA additionally lies near the crest of the equatorial ionization anomaly. In all the stations, the EQ induced anomalies are ~ 5 TECU, whereas the geomagnetic storm anomalies after the main shock are >10 TECU. This shows that ionospheric responses to the storm ($Dst = -150$ nT

and $Kp \geq 7$) are more than EQ anomalies in the analysis of all IGS GNSS stations. However, EQ induced anomalies also exist and it cannot be denied.

In addition to the temporal TEC analysis, seismo ionospheric anomalies are also investigated before and after the main shock in the spatial format of TEC in the 4 GNSS stations within the critical region. Like temporal TEC, spatial values also exhibit significantly low intensity ionospheric anomalies before and after the main shock (Figs. 7). The AUCK and OUS2 stations also detect low TEC anomalies of < 5 TECU associated with the EQ which may be to the low latitude of these GNSS stations. Figure 8 shows the diurnal TEC and differential TEC for the KOUC and ASPA stations. An interesting ionospheric pattern occurred before the main shock as detected by the ASPA station (Fig. 8; Panel, d). A low intensity TEC perturbation began at LT=10:00 on the 10th days before the main shock, and peaked at 5-6 TECU during LT=16:00-18:00 on the 6th day before the main shock. Furthermore, this anomaly attenuates with minimum values of < 5 TECU on the day before the seismic activity (Fig. 8). This enhancement and depletion of the TEC anomaly is due to the proximity of the ASPA station to the seismogenic zone. This is also evidence that EQ energy during the preparation period effects the closest stations. It also provides motivation to enhance the cluster of GNSS stations within the world's most severe seismic zones. The mutual enhancement and depletion of low TEC anomalies during EQ preparation period endorses the hypothesis of lithosphere ionosphere coupling. The time series of diurnal TEC and dTEC manifest anomalous values consequent to the EQ and geomagnetic storm for more than 10 TECU in all GNSS stations. The anomalous values in spatial analysis are about 20 – 30% beyond the confidence bounds and the variations began at LT=10:00 on the same day. Moreover, the ionospheric anomalies lingered over the epicenter till LT=18:00 during the EQ preparation period, after which the anomalies ceased and the M_w 8.2, Fiji event occurred. On the other hand, the

geomagnetic storm induced high intensity TEC anomalies of >15 TECU occurred at LT=12:00-22:00 on the 7th days after the main shock. Ionospheric perturbations due to the geomagnetic storm are unanimously detected by all four GNSS stations around the epicenter of the future EQ. This evidences that a large majority of strong ionospheric anomalies are induced by the geomagnetic storms; therefore, careful analysis of geomagnetic storms are mandatory in order to distinguish between EQ induced or storm induced TEC anomalies.

In this paper, we report TEC anomalies from GNSS stations of low and high intensity associated with an EQ and geomagnetic storm, respectively. These analyses clearly differentiate the seismic and non-seismic ionospheric anomalies and we further emphasize the necessity of further work in this field with multiple satellites and GNSS measurements. Previously, many reports investigated the relationship of lithosphere and ionosphere using GNSS and other ionosphere monitoring satellites in equatorial and low to high latitudinal regions. For example, Pulinets and Ouzounov (2011) presented the formation of ionospheric anomalies as resulting from continuous emission of Radon from the EQ breeding zone during the preparation period. Moreover, the concept of p- hole emission from squeezed rocks due to tectonic forces also cause variation in nearby air over epicenter and subsequently perturb the ionosphere (Freund, 2009). Shah et al. (2019) investigated ionospheric anomalies by differential electron density values in the Chilean region by comparing equatorial and low latitudinal electron density. Similarly, Kuo et al. (2011) demonstrated the coupling of lithosphere and ionosphere over the epicenter due to squeeze rocks from tectonic stress under a controlled environment through numerical simulation. Furthermore, they showed the potential difference of the electric field around the epicenter of the future EQ and also traced the initial and terminal points of the ionospheric anomaly. The results in this paper also present sufficient evidence for ionospheric anomalies associated with EQ. In contrast to other studies, we

find significantly higher intensity TEC anomalies induced by a geomagnetic storm as compared to the lower intensity seismic anomalies.

4. Conclusions

In this paper, temporal and spatial TEC from GNSS stations for 20 days before and 10 days after the M_w 8.2, Fiji EQ are investigated for seismo ionospheric anomalies. However, a severe geomagnetic storm ($Dst = -150$ nT and $Kp \geq 7$) also causes ionospheric anomalies within 7-10 days after the main shock. Low intensity EQ anomalies appear within 5-10 days before and after the main shock during quiet storm conditions ($Dst > -50$ nT and $Kp \leq 3$). The severe geomagnetic storm ($Dst = -150$ nT and $Kp \geq 7$) causes intensive TEC anomalies beyond 5 days after the main shock. Both dTEC and % TEC show more ionospheric variations related to the geomagnetic storm as compared to the EQ. Spatial TEC series show prominent anomalies associated with the EQ during LT=10:00-18:00 on abnormal days. On the other hand, geomagnetic storm anomalies of high intensity > 10 TECU were detected at LT=12:00-22:00 on the 7th day after the main shock at all GNSS stations. These anomalies can be explained by the hypothesis of lithosphere ionosphere coupling; however, more analysis with measurements from future GNSS stations purposefully clustered in severe seismic zones will play a decisive role in understanding the coupling between the lithosphere and ionosphere.

Authors statement

MS designed the study. AK and SH performed analysis. MAA helped in plots. AK, SH, MS wrote manuscript. HHS and NAN gave final remarks to reviewers.

Conflict of interest

The authors declare no conflict of interest with other authors and no conflict of interest with any funding organization.

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Table 1: Details about EQ and Geomagnetic storm anomalies from GNSS stations

Station	Station code	Geographical Location	Distance from epicenter (km)	EQ anomaly			Storm Anomaly		
				Day	dTEC	% TEC	Day	dTEC	% TEC
Dunedin (New Zealand)	OUS2	45°S,170°E	2971	-3	-1.9	-60	7	4	90
				-2	2.3	40	8	1.8	60
Whangaparaoa (New Zealand)	AUCK	36.6°S, 174.7° E	1887	-6	4	43	7	-1.9	-60
							8	2.3	40
Koumac (New Caledonia)	KOUC	20.5°S,164.2°E	1771	-1	5	60	7	14	150
				2	3	30			
Pago Pago (USA)	ASPA	14°S,170° W	1104	-8	5	30	7	14	100
				-7	5	30	9	12	50
				-6	7	40	10	6	20
				-1	-2	-20			
				2	6	30			

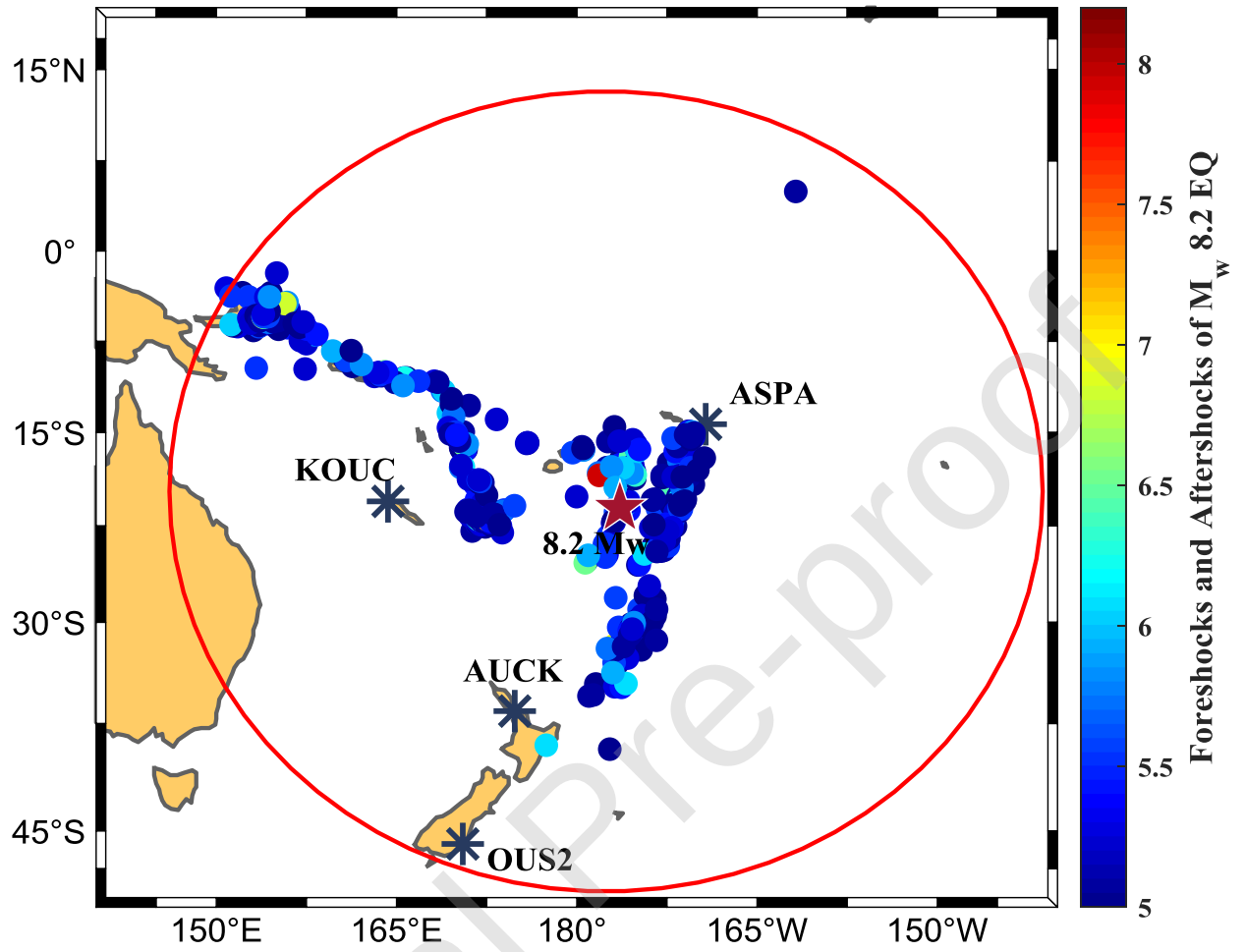


Figure 1. Geographical locations of the IGS GNSS stations and epicenter of the EQ. The red circle shows the Dobrovolsky region around the epicenter.

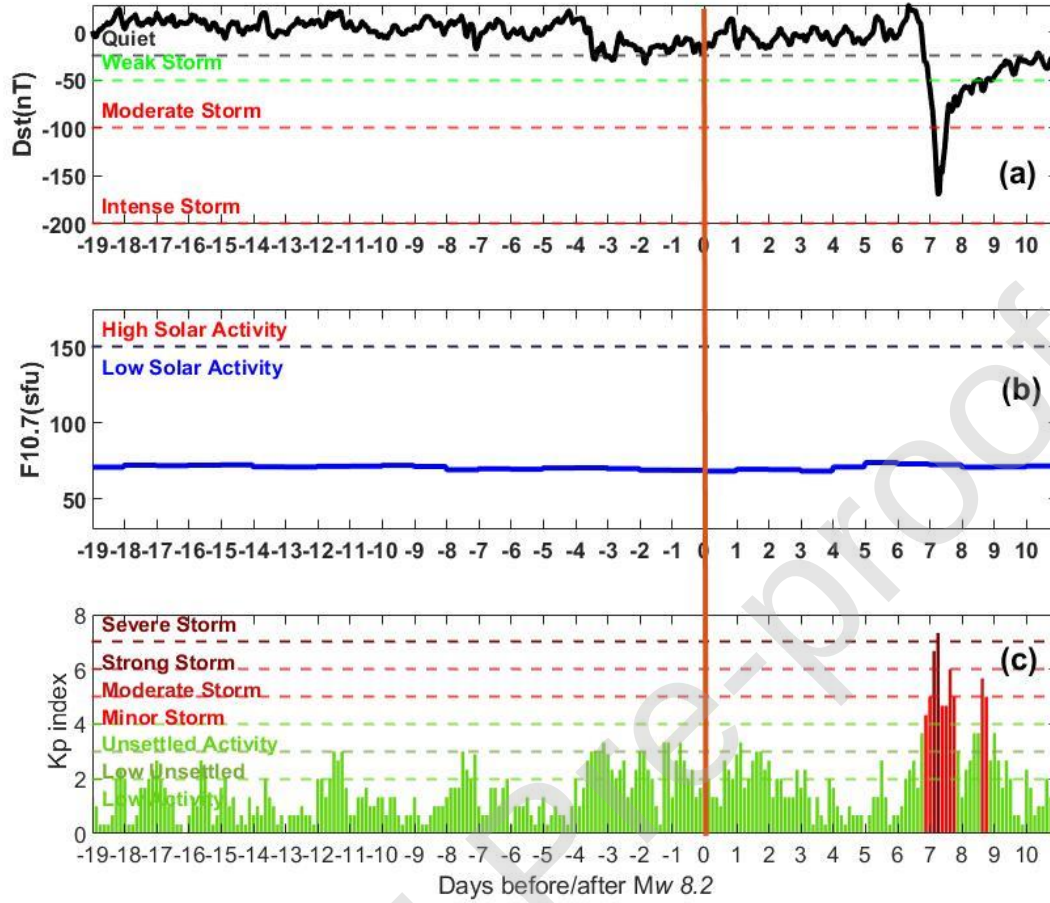


Figure 2. Geomagnetic storm conditions during the observational period. (a) Dst, whereas (b) and (c) represent F10.7 and Kp indices, respectively.

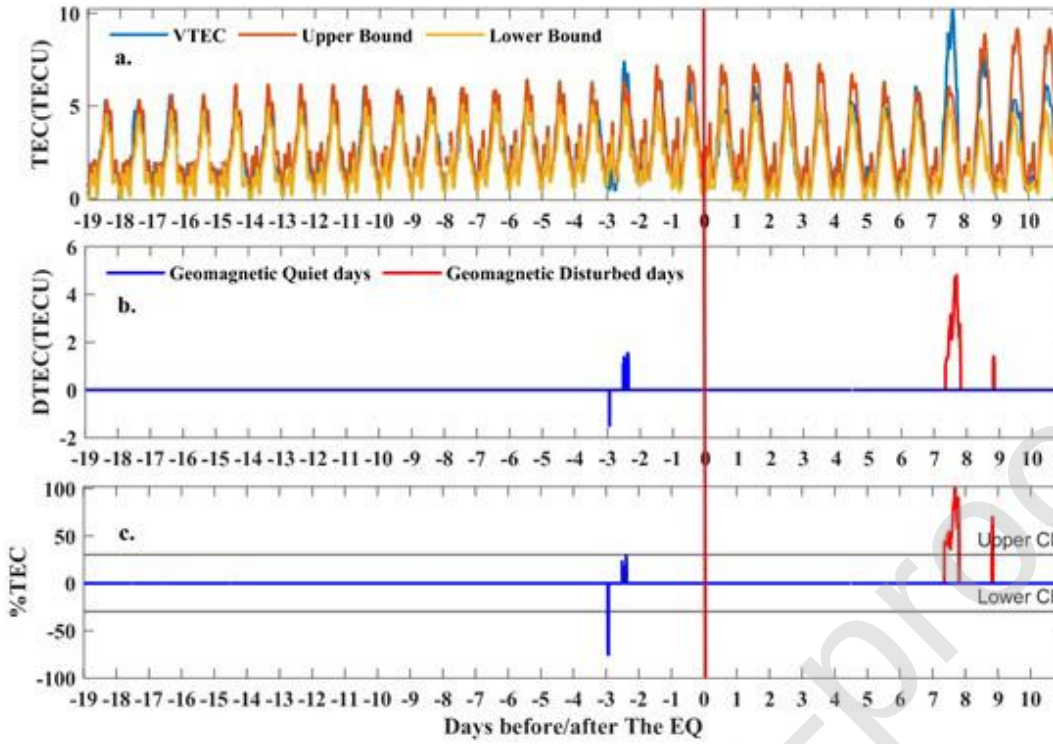


Figure 3. Temporal TEC variations with upper and lower confidence bounds for OUS2 station. (a) shows original TEC by blue line whereas orange and yellow lines represents the associated upper and lower bounds. (b) Shows the TEC deviations from upper bound and (c) shows the percentage deviation in TEC from upper and lower bounds.

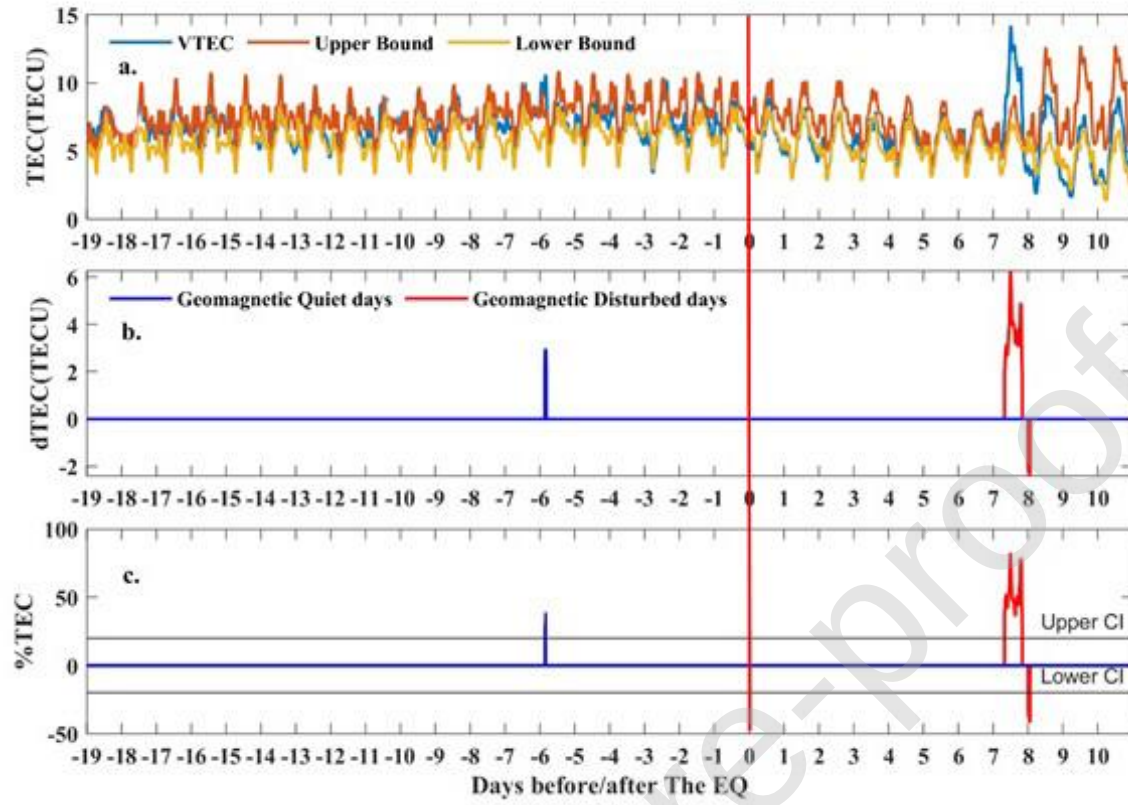


Figure 4. Original TEC, dTEC and % TEC with upper and lower bounds at AUCK stations for analysis of ionospheric anomalies related to the Fiji EQ.

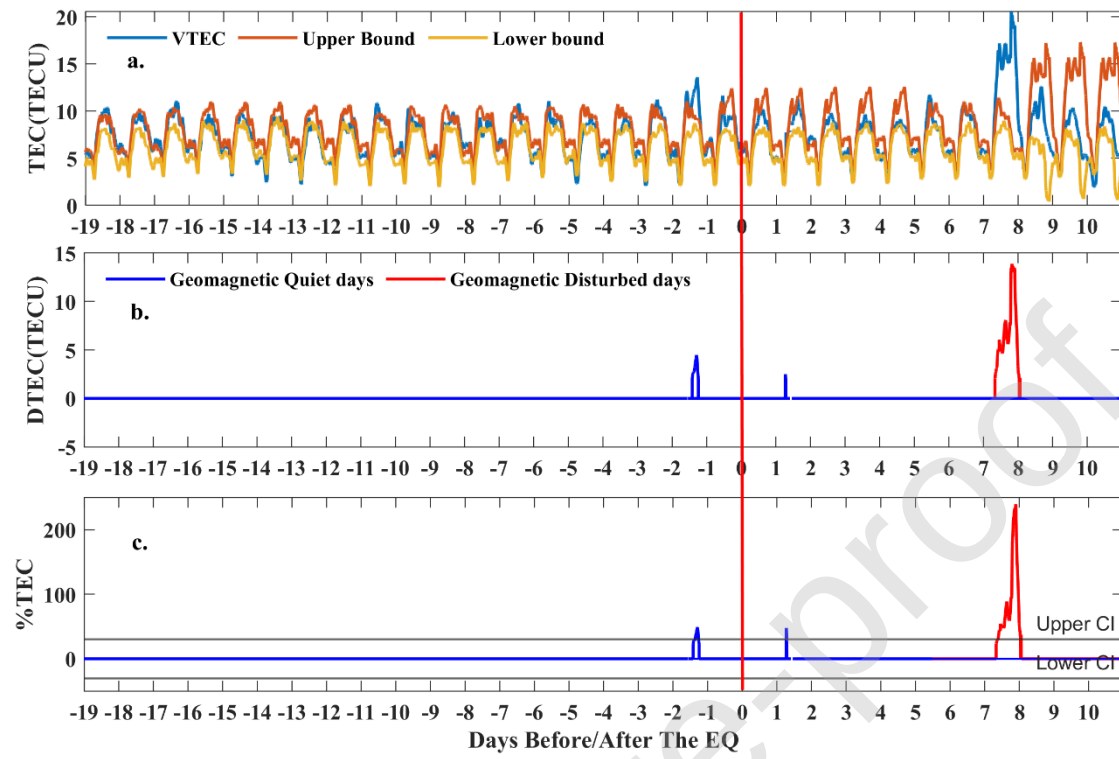


Figure 5: TEC, dTEC and % TEC with associated upper and lower bounds from KOUC station.

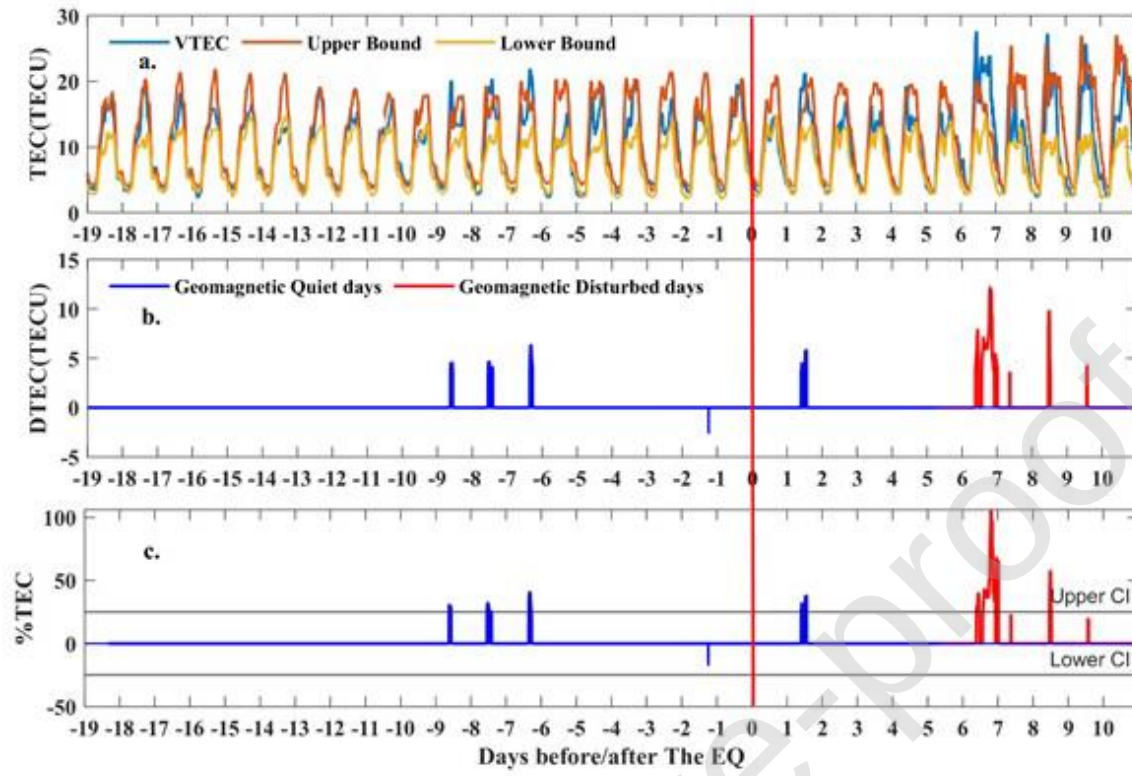


Figure 6: ASPA station's TEC, dTEC and percentage TEC for 20 days before and 10 days after the main shock.

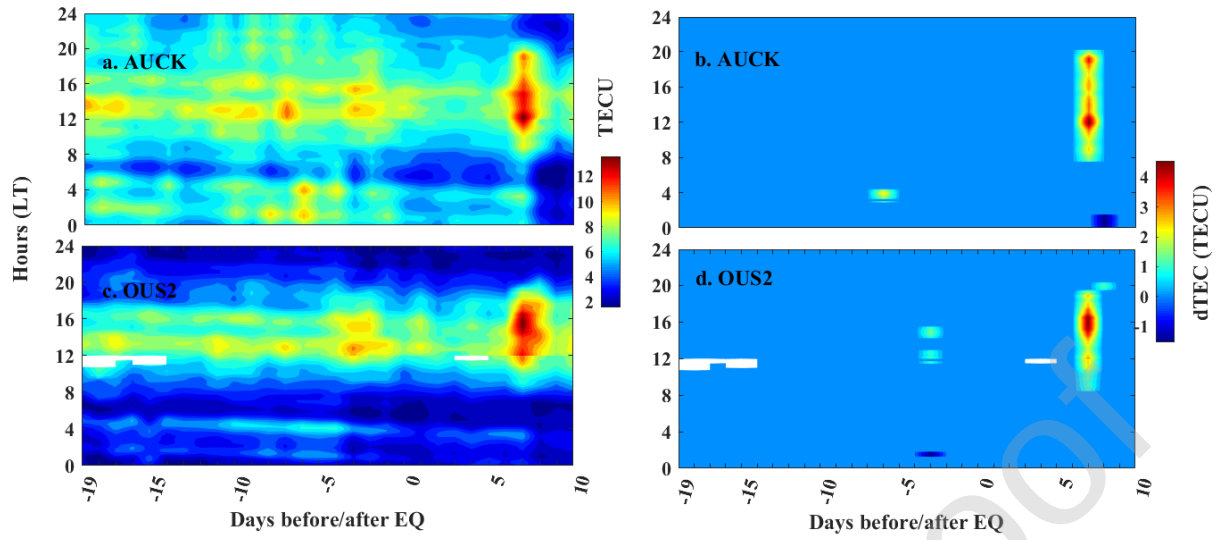


Figure 7: Temporal TEC and dTEC for AUCK and OUS2 stations for 20 days before and 10 days after the main shock.

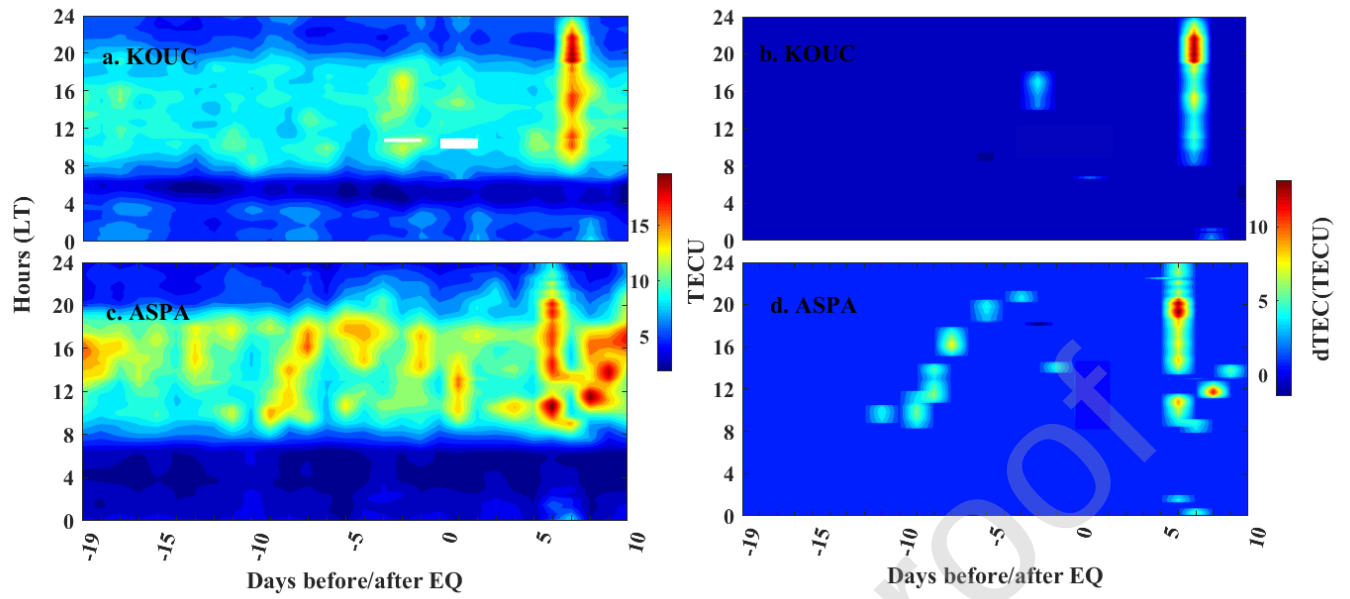


Figure 8: Temporal analysis for ionospheric anomalies associated with Fiji's EQ and geomagnetic storm from KOUC and ASPA stations.